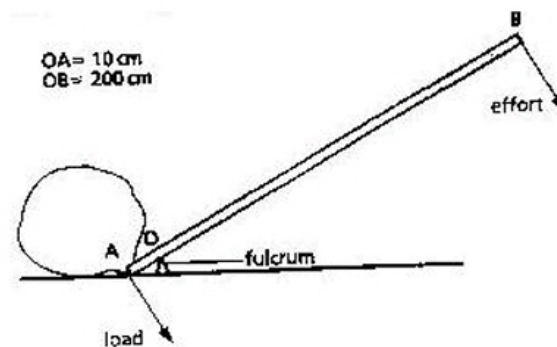


SIMPLE MACHINES

- A machine is a device that helps us to do work easier e.g. levers, pulleys, inclined planes, gears, wheel and axle.
- A machine is anything that magnifies a small force into a large force that is able to do work.
- Machines are energy convertors

Levers



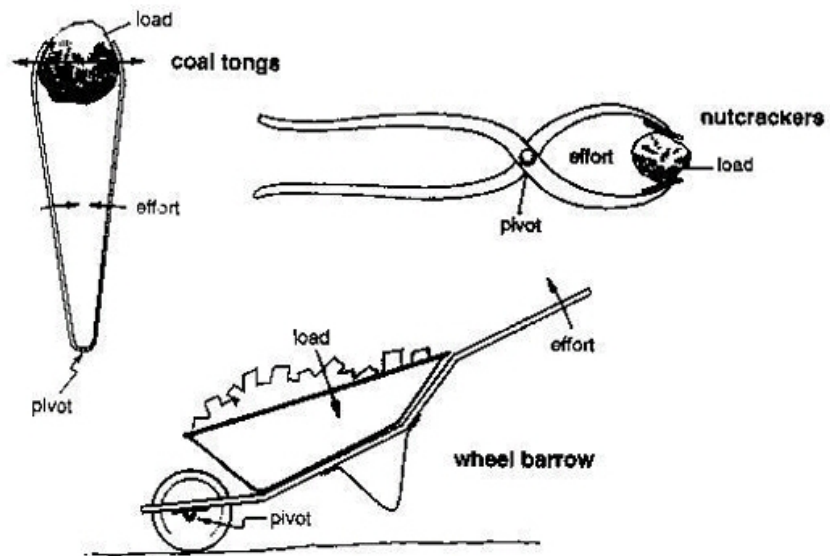
- A lever is a device which can turn about a pivot

21 | Prepared by Farirayi D.T / 0736243049

- In a working lever a force called the effort is used to overcome a resisting force called the load. The pivotal point is called the fulcrum
- If we use a crow bar to move a heavy boulder, our hands apply the effort at one end of the bar and the load is the force exerted by the boulder on the other end.
- A lever changes the force applied on one end of the lever to an opposite force on the other side

Classes of lever

- First class: fulcrum is between the load and effort e.g. crowbars, scissors, claw hammer, pliers.
- Second class: load is between the fulcrum and effort e.g. wheelbarrow, nutcracker, bottle opener
- Third class levers: effort is between the fulcrum and load e.g. a hoe, fishing rod, tongs, a spade



MACHINES

- A machine is a device that causes a change in the way that these forces act.
- Machines are mechanical systems designed to make hard work easier e.g. inclined planes, gears, pulleys and levers by using less applied force (effort) to lift a larger force (load)

Mechanical Advantage (MA)

- It is the measure of how much easier it is to complete a task using a simple machine. The greater the MA, the easier it is to do work
- Is the ratio of two forces

$$MA = \frac{\text{load (L)}}{\text{effort (E)}}$$

Velocity Ratio (VR)

- Is the ratio of two distances i.e. distance moved by the effort and the distance moved by load.

$$VR = \frac{\text{effort distance}}{\text{load distance}}$$

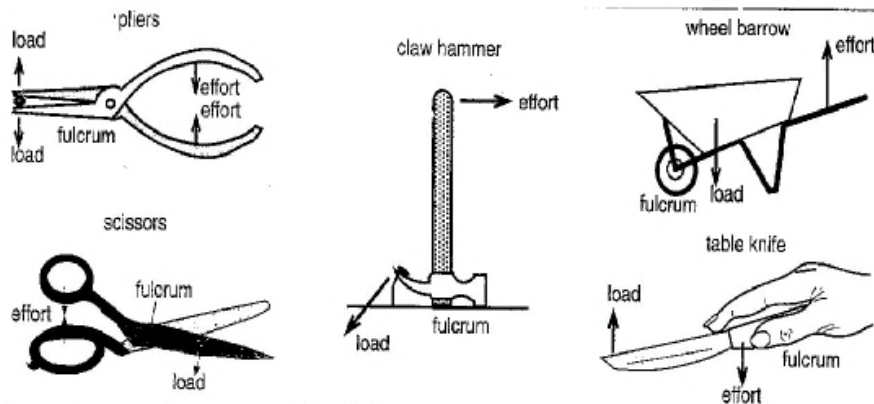
Efficiency

- Is the ratio of work output (MA) and work input (VR). It can be expressed as a percentage.

$$\text{Efficiency} = \frac{MA}{VR} (\times 100\%)$$

NB Efficiency, VR and MA are all ratios hence have no units

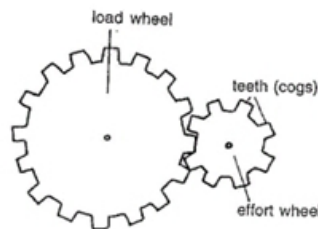
Levers



- Is a rigid bar that rotates around a fixed support called a fulcrum to lift a load
- The lever is a machine which acts as a force multiplier i.e. it allows a smaller force (effort) to overcome a large force (load) e.g. pliers, crowbar, scissors, wheelbarrow
- By increasing the effort distance MA and VR increases while efficiency remains the same.
- Efficiency is improved through lubrication as it reduces friction

Gears

- A gear is a system of wheels with interlocking teeth so that one wheel drives another by transferring force from one gear to another.
- Effort gear is smaller than load gear and has to be turned several times for the load gear to turn



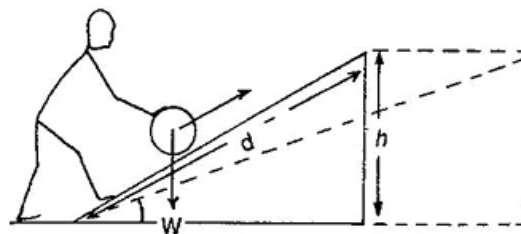
- A gear with 12 teeth makes two revolutions to drive and rotate another gear with 24 teeth a full cycle.

$$VR = \frac{\text{number of teeth on load gear}}{\text{number of teeth on effort gear}}$$

$$MA = \frac{\text{diameter of load gear}}{\text{diameter of effort gear}}$$

- Efficiency can be improved by
 - Lubrication
 - Appropriate shaping of gear teeth
 - Proper alignment of gear teeth to reduce friction

Inclined Plane



- Is a flat surface that is at an angle to the horizontal

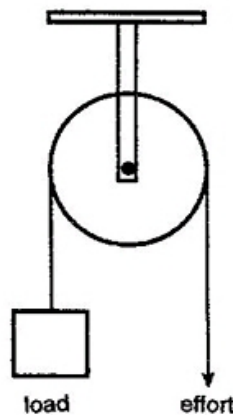
- Efficiency can be improved through
 - Lubrication
 - Increasing length of incline

$$VR = \frac{\text{length of incline}(d)}{\text{height of incline}(h)}$$

Pulleys

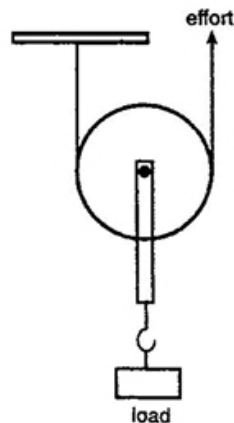
- Is a wheel with a groove through which a rope or belt passes

Single fixed pulley



- The load is raised off the ground by an effort which is also directed down wards; therefore the pulley changes the direction of the force.
- $MA = 1$ because the values of the load and effort will be the same
- $VR = 1$ because distance moved by the load and effort must be the same

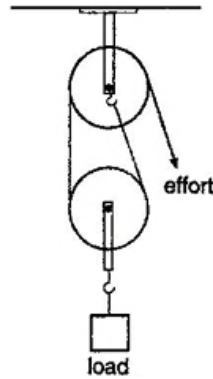
Single moveable pulley



- The tension in the string is the same as the effort i.e. It is also an upward force.

- $MA = 2$ because the load is two times the effort
- $VR = 2$ because as the pulley moves up the load, the string moves twice the distance.

Block and Tackle



- Two or more pulleys used. The machines become a force multiplier as well as direction changer.
- It uses both fixed and moveable pulley
- $VR =$ the number of pulleys/ropes

Efficiency of pulleys can be improved through

- Lubrication
- Using lighter pulleys
- Increasing number of pulleys

Energy losses in machines

- Machines are used to do work, converting energy from one form to another. In practice machines are never 100% efficient. This means that total energy input is greater than the useful work output.
- Some of the energy input is converted to unwanted forms such as heat and sound. To overcome or minimise friction, the movable parts of the machine are lubricated or using rollers and ball bearings.
- Some of the energy input is used to overcome mass of moving parts of the machine hence there is need for mass reduction.